



M.C.A. (Two Years Course) First Year First Semester (CBCS)
IT4 Paper-IV - Advanced DBMS and Administration

P. Pages : 2

Time : Three Hours



KRS/KS/25/1781

Max. Marks : 80

- Notes : 1. All questions are compulsory and carry equal marks.
 2. Draw well labelled diagram wherever necessary.

Either :

1. a) What is normalization? Explain 4NF with suitable example. 8
- b) Explain the structure of query optimizer with suitable diagram. 8

OR

- c) Write note on followings: 8
 - i) Merge Join
 - ii) Hash Join
- d) What is Equivalence of Expressions? Explain different equivalence rules. 8

Either :

2. a) Explain the followings: 8
 - i) Shadow paging
 - ii) Check points
- b) Explain database environment for client server computing. 8

OR

- c) What is concurrency control and time stamp ordering? Explain. 8
- d) Write note on: 8
 - i) Crash recovery techniques
 - ii) States of transactions.

Either :

3. a) Explain Oracle Export/Import utilities in details. 8
- b) Write note on: 8
 - i) Control files
 - ii) Redo log files.

OR

- c) Explain Oracle database architecture with proper diagram. 8

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- d) How data integrity and password security managing by Oracle database administrator. 8

Either :

4. a) Explain process of database configuration and I/O issues in detail. 8
b) Explain Oracle performance tuning methodology in detail. 8

OR

- c) Explain roll back segment tuning in detail. 8
d) Write note on: 8
i) Redo log buffer ii) Buffer cache
5. a) Explain lossless join decomposition with one example. 4
b) Explain ACID properties of transactions. 4
c) Explain Privileges and role of user in brief. 4
d) Explain Oracle alert and trace files. 4
